Aryasatya Muhammad Aqsel

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PROFILE SUMMARY

Fresh Graduate Informatics Engineering from University of Dian Nuswantoro (GPA 3.48/4.00) with expertise in Data Analysis, Machine Learning, and Data Visualization. BNSP Certified Associate Data Scientist with experience in Python, Scikit-learn, XGBoost, and interactive dashboard development using React and Streamlit. Published researcher in the Sinta 3 accredited Journal of Applied Intelligent Computing (JAIC, 2025). Passionate about transforming data into actionable insights and building data-driven solutions.

EDUCATION

University of Dian Nuswantoro — **Semarang, Indonesia**
Bachelor of Informatics Engineering — 2022 - 2026

- **GPA:** 3.48 / 4.0
- **Honors:** Very Satisfactory
- **Relevant Coursework:** Machine Learning, Data Mining, Statistics, Computer Vision, Data Analyst, Data Science

WORK & LEADERSHIP EXPERIENCE

Department of Communications and Information Technology — **Semarang, Indonesia**
Intern, Software Development Team — May 2025 - Jun 2025

- Contributed to a team of 4 in developing a Laravel-based intern activity reporting system, engaging in cross-functional collaboration with stakeholders to translate manual tracking requirements into an automated digital solutionConducted functional testing and bug identification across system features to ensure reliability prior to deployment
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- Monitored backend performance and assisted in MySQL database oversight and SQL query optimization to maintain data integrity

SKILLS, ACTIVITIES & INTERESTS

Languages: Native in Bahasa Indonesia; Professional in English
Technical Skills: Python, TensorFlow, SQL, Microsoft Excel, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn
Certifications & Training:

- Associate Data Scientist BNSP / LSP UDINUS (Apr 2026)
- Foundations: Data, Data, Everywhere – Google Coursera (Jun 2026)
- AI Literacy for Everyone BASS Training Center × Google.org & ADB (Nov 2025)
- UX Writing Short Class MySkill (Oct 2024)

Interests: Formula 1 data engineering and race strategy analytics; environmental data science and causal inference; data analyst

PROJECTS & RESEARCH

Gue Ngekost Offline-First Personal-Finance PWA for Boarding-House Residents | *React, Vite, Zustand, Tailwind CSS, Recharts, PWA*

- Built an installable, fully offline Progressive Web App (React 18 + Vite 5) spanning 6 modules cashflow, food tracker, packing list, analytics with 100% client-side localStorage persistence and zero backend, sign-up, or telemetry
- Shipped a bilingual (ID/EN) interface with one-tap locale switching that propagates to every label, chart, and PDF export, plus an adaptive layout scaling from a 320 px phone tab bar to a desktop sidebar rail
- Engineered analytics with 6-month Recharts visualizations and configurable jsPDF reports (per month / 6 months / year), and hardened reliability via a lazyWithRetry chunk-recovery loader + service-worker cache control, eliminating post-deploy blank-screen failures

F1 Race Strategy Intelligence Dashboard | *Python, FastF1, XGBoost, SHAP, Streamlit, Next.js*

- Built dual-frontend F1 analytics platform covering 2022-2024 seasons: telemetry, tyre strategy, and driver head-to-head comparisons; deployed on Streamlit Cloud and Vercel
- Developed ML race-outcome classifier (DNF / Podium / Points / Outside Points) with SHAP explainability integrated into interactive dashboard
- Engineered modular data pipeline from FastF1 and Ergast APIs with structured feature engineering for ML and visualization

Coffee Bean Classification Using ResNet50 | *Python, TensorFlow/Keras, OpenCV*

- Developed two-stage transfer learning + fine-tuning pipeline to classify 4 Indonesian coffee bean varieties (Arabica, Robusta, Excelsa, Liberica) across 2,000 images
- Achieved 94.5% test accuracy; primary confusion source identified as Excelsa Arabica similarity (72.7% of errors), documented in analysis
- Published in JAIC (SINTA 3 indexed)

Interactive Portfolio Website | *Next.js, React, TypeScript, Tailwind CSS, Framer Motion, GSAP*

- Designed and built a single-page animated portfolio for a graphic designer, featuring a custom cursor, magnetic
- Engineered scroll-driven motion using Lenis smooth scroll, Framer Motion reveal animations, and GSAP parallax for a fluid, app-like browsing experience
- Built on Next.js App Router with React Server Components, next/font, and OpenGraph metadata for fast load and SEO; fully responsive across devices

PaperStory Interactive Research Scrollytelling Platform | *Next.js, React, TypeScript, Tailwind CSS, Framer Motion, Vercel*

- Built a web platform that reimagines academic papers as scroll-driven visual narratives, transforming 7 published papers (computer vision, FHE hardware, motorsport ML, environmental law, public health, urban policy) into 70 custom-animated scenes
- Engineered a data-driven, scene-based architecture a single source-of-truth story registry rendering 10 modular scenes per paper through reusable Framer Motion visualization components, fully responsive down to mobile.